

Supplemental Appendix: Cleansing Oil

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A Empirical Validation of Oil Shock Identification

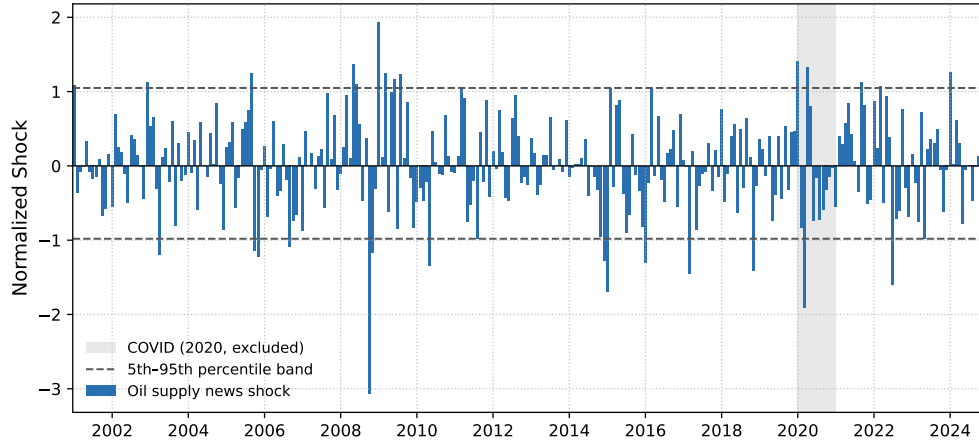
Figure A.1 plots the oil supply news shock of Känzig (2021) over my sample, together with the 5th–95th percentile band used in the trimmed-shock robustness checks.

An alternative set of oil supply shocks is that of Baumeister and Hamilton (2019), identified from a sign-restricted Bayesian structural vector autoregression designed to disentangle oil supply from oil demand disturbances. Over my sample, however, they yield a response of industrial production that contradicts both theory and the empirical literature — potentially failing to cleanly separate supply from demand effects, which renders them unsuitable for this analysis.

Figure A.2 compares the impulse response functions generated from estimating equation (3) for the real WTI oil price and OECD+6 industrial production index using both the oil news shock of Känzig (2021) and the structural oil supply shock of Baumeister and Hamilton (2019), both normalised to a 10% increase in oil prices on impact. The two series generate virtually identical price dynamics: oil prices rise sharply on impact and decay to zero within 8 months, so agents face the same price signal regardless of which shock is used. The two series diverge, however, in their implied economic activity effects. The news shock delivers the expected contractionary response: industrial production declines after a few months and remains depressed through the 2-year horizon. The structural supply shock, by contrast, yields a positive and statistically significant increase in industrial production over the 2001–2024 sample — the opposite of what theory and the broader literature predict — regardless of the controls included.

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Figure A.1: Oil Supply News Shock

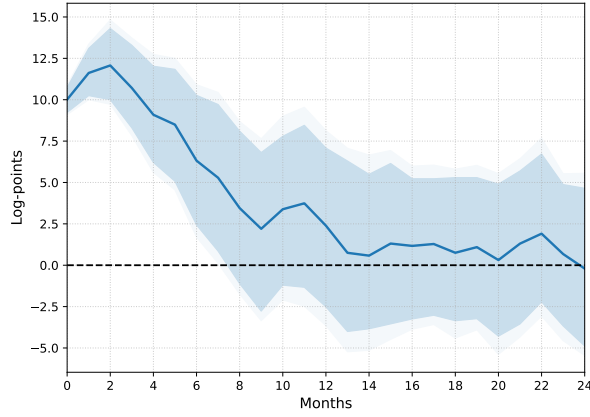


Notes: Monthly oil supply news shock of Känzig (2021), normalised so that a 1-unit shock raises oil prices by 10% on impact. Dashed lines mark the 5th and 95th percentiles of the shock over 2001–2025; shocks outside this band are excluded in the trimmed-shock robustness checks. The shaded area marks 2020, excluded from the estimation. Shocks: January 2001 – December 2024.

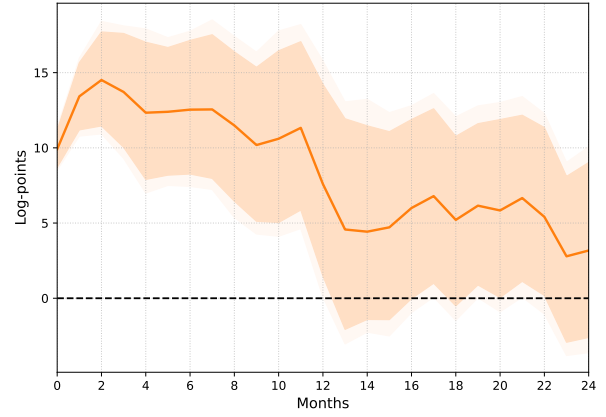
There are many reasons why this could be happening. One is the contamination of the shocks, as argued by Mori and Peersman (2024). Nevertheless, including more financial controls does not solve the problem. When using a shorter sample starting in 2010, consistent with the panel data, the problem becomes less severe, though the response is more delayed than what has been documented in the literature. The results are plotted in figure A.3. The oil news shock, by contrast, generates coherent dynamics across the samples used in the main analysis: starting in 2001, when the emissions sample begins, or in 2010, when panel coverage broadens significantly, the shock induces the same response in industrial production and oil prices, albeit with different persistence. I therefore use the oil news shock throughout to ensure consistency.

I also compare the dynamics of the other macroeconomic indicators used in subsection 4.2 — the 10-year nominal interest rate, the term spread, macroeconomic uncertainty, the geopolitical risk index, and oil price uncertainty (OVX) — induced by the oil news shock and the structural oil supply shock. Figure A.4 shows that, unlike industrial production, the induced impulse response functions are similar across both shocks. In particular, oil price uncertainty falls in the first months after both shocks. Because the structural supply shocks of Baumeister and Hamilton (2019) are not identified from OPEC announcements, this decline does not merely reflect the uncertainty that such announcements resolve.

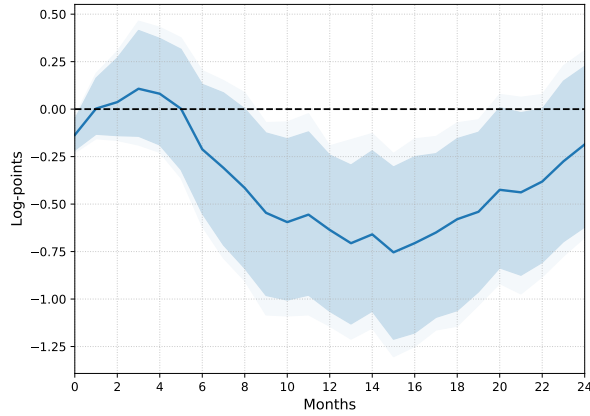
Figure A.2: Oil News Shock vs. Structural Supply Shock: Price and Activity Responses



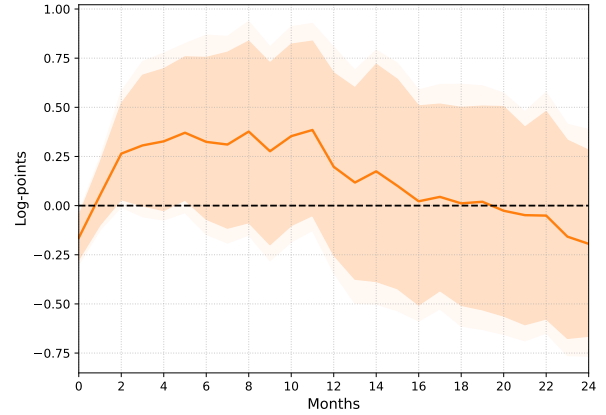
(a) Real WTI — Oil News Shock



(b) Real WTI — Structural Supply Shock



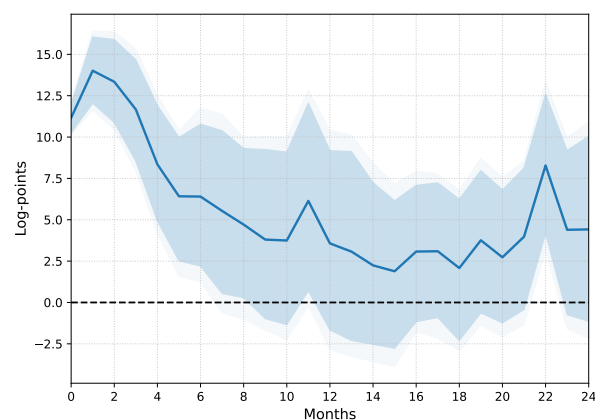
(c) Industrial Production — Oil News Shock



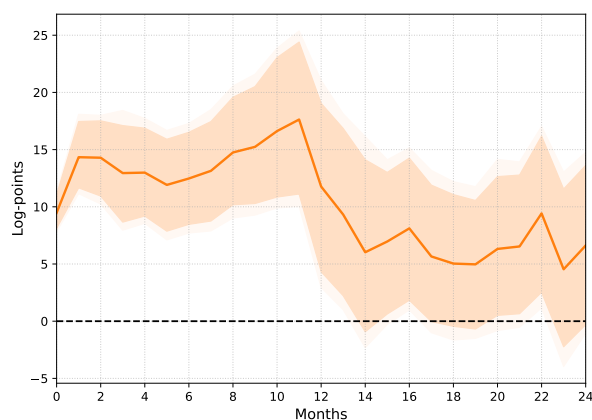
(d) Industrial Production — Structural Supply Shock

Notes: Impulse responses to each shock series, normalised to a 10% increase in oil prices on impact. Left column: oil supply news shock of Känzig (2021). Right column: structural oil supply shock of Baumeister and Hamilton (2019). Top row: real WTI oil price (log-points). Bottom row: OECD+6 industrial production (log-points). Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

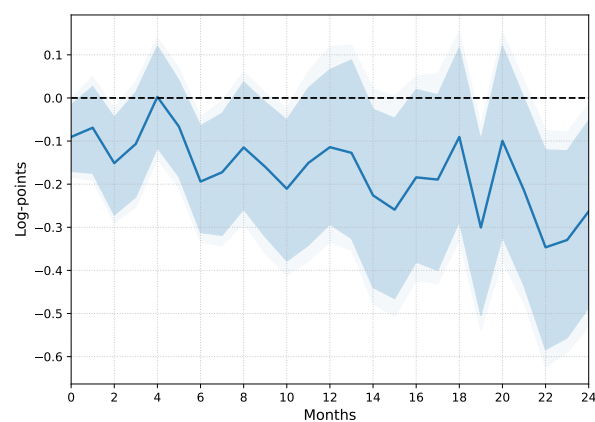
Figure A.3: Oil News Shock vs. Structural Supply Shock: Price and Activity Responses — 2010 Sample



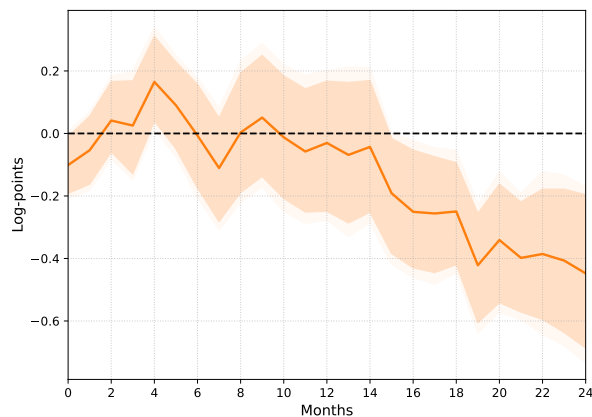
(a) Real WTI — Oil News Shock



(b) Real WTI — Structural Supply Shock



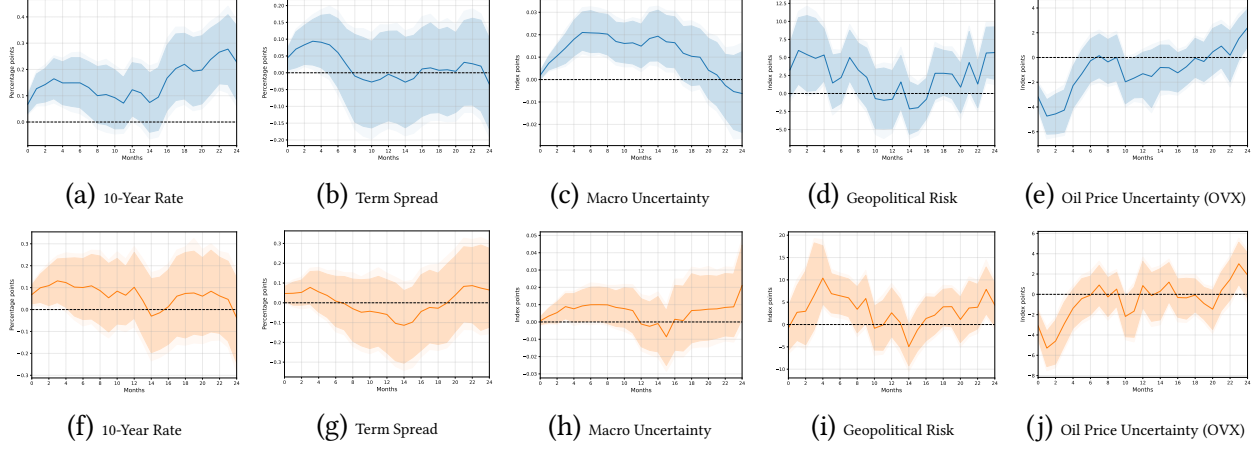
(c) Industrial Production — Oil News Shock



(d) Industrial Production — Structural Supply Shock

Notes: Impulse responses to each shock series, normalised to a 10% increase in oil prices on impact. Left column: oil supply news shock of Känzig (2021). Right column: structural oil supply shock of Baumeister and Hamilton (2019). Top row: real WTI oil price (log-points). Bottom row: OECD+6 industrial production (log-points). Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded. Shocks: January 2010 – December 2024; outcome data extends to December 2025.

Figure A.4: Oil News Shock vs. Structural Supply Shock: Financial and Uncertainty Responses



Notes: Impulse responses to each shock series, normalised to a 10% increase in oil prices on impact. Top row: oil supply news shock of Känzig (2021). Bottom row: structural oil supply shock of Baumeister and Hamilton (2019). Oil price uncertainty is the CBOE crude oil volatility index (OVX). Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded. Shocks: January 2002 – December 2024, except the OVX panels, which start in June 2008 because the index is only available from June 2007; outcome data extends to December 2025.

B Construction of the CO_2 -Equivalent Concentration Index

The time-series analysis uses a CO_2 -equivalent (CO_2e) concentration index that aggregates the atmospheric burden of 4 major greenhouse gases— CO_2 , CH_4 , N_2O , and SF_6 —into a single, consistent metric. Monthly global-mean concentrations of CO_2 (ppm), CH_4 (ppb), N_2O (ppb), and SF_6 (ppt) are obtained from NOAA.

Radiative forcing. Each gas contributes to total radiative forcing (RF) relative to its pre-industrial (1750) reference concentration. I use the simplified effective radiative forcing (ERF) expressions of Etminan et al. (2016), as updated by Meinshausen et al. (2020) and adopted in Intergovernmental Panel on Climate Change (IPCC) (2023b) and in NOAA’s Annual Greenhouse Gas Index (NOAA Global Monitoring Laboratory 2025). Let C , M , N , S denote current concentrations of CO_2 (ppm), CH_4 (ppb), N_2O (ppb), and SF_6 (ppt), with pre-industrial (1750) reference values $C_0 = 278.3$, $M_0 = 729.2$, $N_0 = 270.1$, $S_0 = 0$ (Intergovernmental Panel on Climate Change

(IPCC) 2023a). The individual ERF terms (W m^{-2}) are:

$$\Delta F_{\text{CO}_2}(C, N) = \left[\alpha(C) + c_1 \sqrt{N} \right] \ln\left(\frac{C}{C_0}\right) \cdot e_1, \quad \alpha(C) = d_1 + a_1(C - C_0)^2 + b_1(C - C_0) \quad (1)$$

$$\Delta F_{\text{CH}_4}(M, N) = \left[a_2 \sqrt{M} + b_2 \sqrt{N} + d_2 \right] \left(\sqrt{M} - \sqrt{M_0} \right) \cdot e_2 \quad (2)$$

$$\Delta F_{\text{N}_2\text{O}}(C, M, N) = \left[a_3 \sqrt{C} + b_3 \sqrt{N} + c_3 \sqrt{M} + d_3 \right] \left(\sqrt{N} - \sqrt{N_0} \right) \cdot e_3 \quad (3)$$

$$\Delta F_{\text{SF}_6}(S) = \omega (S - S_0) \quad (4)$$

The scalars e_1 , e_2 , e_3 convert the stratospheric-temperature-adjusted forcing into effective radiative forcing, which also accounts for rapid adjustments in tropospheric temperature, water vapour and clouds. Total forcing is $\Delta F_{\text{total}} = \sum_g \Delta F_g$. Parameter values are listed in table B.1.

Table B.1: ERF parameter values (Meinshausen et al. 2020; Intergovernmental Panel on Climate Change (IPCC) 2023b; NOAA Global Monitoring Laboratory 2025). [†]Units of $\text{W m}^{-2} \text{ppt}^{-1}$.

Gas	Parameter	Value
CO_2	a_1	-2.48×10^{-7}
	b_1	7.59×10^{-4}
	c_1	-2.15×10^{-3}
	d_1	5.2488
	e_1 (adj. factor)	1.05
CH_4	a_2	-8.96×10^{-5}
	b_2	-1.25×10^{-4}
	d_2	0.045194
	e_2 (adj. factor)	0.86
N_2O	a_3	-3.42×10^{-4}
	b_3	2.546×10^{-4}
	c_3	-2.44×10^{-4}
	d_3	0.12173
	e_3 (adj. factor)	1.07
SF_6	$\omega^\dagger$	5.67×10^{-4}

CO_2 -equivalent concentration. The CO_2e concentration C_t^{eq} is the unknown in equation (2) of the main text: the level of CO_2 that, with all other gases held at their pre-industrial values, produces the same total forcing as the observed concentrations. In that hypothetical, the CH_4 , N_2O

and SF_6 terms are zero, since each is proportional to the gap between that gas's own concentration and its pre-industrial value $-(\sqrt{M_0} - \sqrt{M_0})$, $(\sqrt{N_0} - \sqrt{N_0})$ and $(S_0 - S_0)$ in equations (2) to (4). The condition therefore reduces to $\Delta F_{CO_2}(C_t^{eq}, N_0) = \Delta F_{total}$, with the CO_2 term still evaluated at N_0 because N_2O enters equation (1) through the overlap of the two gases' absorption bands. Since equation (1) is not analytically invertible in C , C_t^{eq} is solved for numerically each period using Brent's method over the interval $[C_0, 5000]$ ppm.

C Wild Bootstrap Procedure

Confidence bands for the time-series local projections are computed using a residual-based wild bootstrap (Liu 1988). For each horizon h , let $\hat{\beta}_h$ denote the OLS estimate of equation (3) and let $\hat{y}_t$ and $\hat{\varepsilon}_t$ denote the corresponding fitted values and residuals. The procedure is as follows.

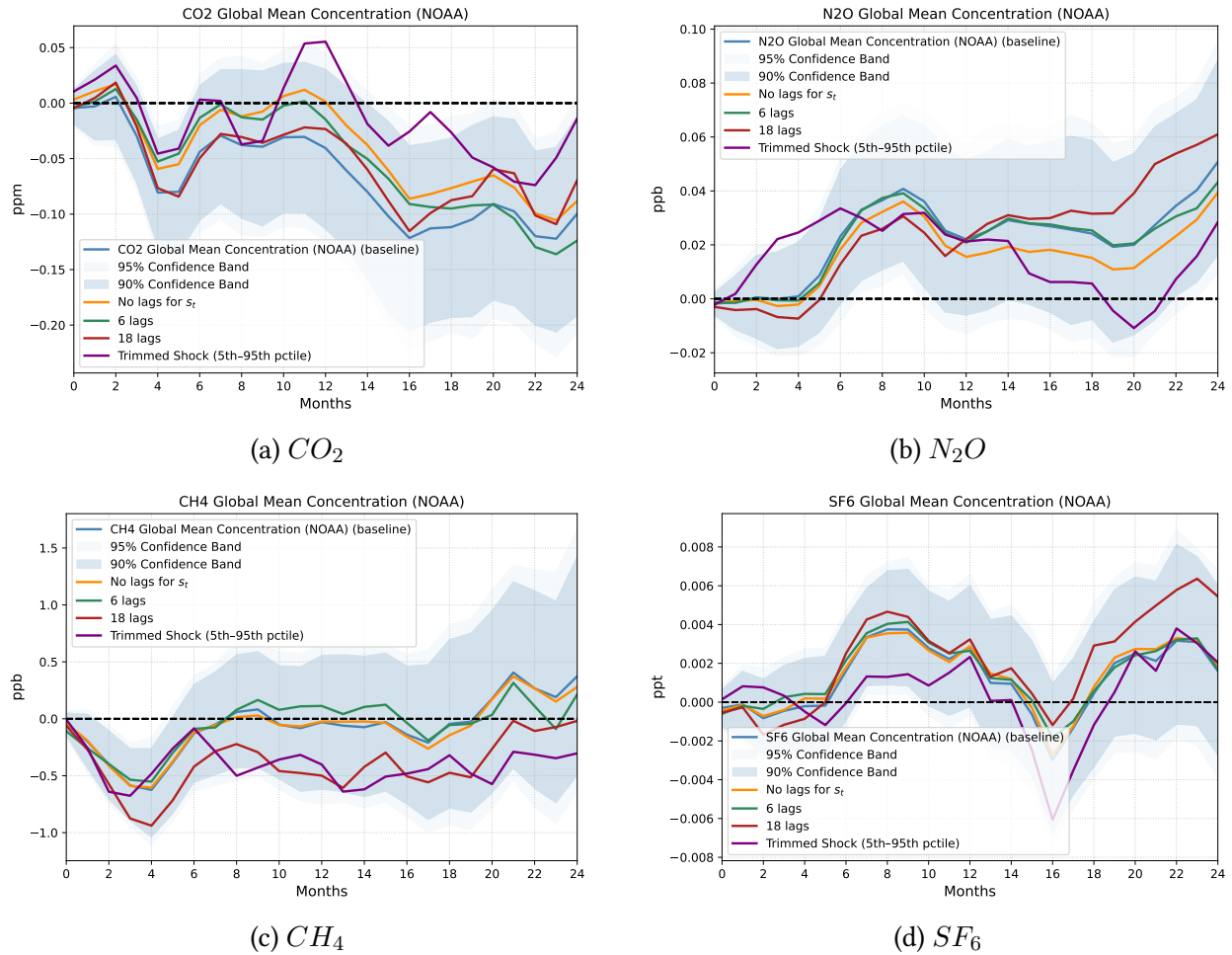
1. For each bootstrap replication $b = 1, \dots, B$, draw an independent multiplier $t_i^{(b)}$ for every observation i from the two-point distribution described below, and construct a bootstrap sample $y_i^{*(b)} = \hat{y}_i + t_i^{(b)} \hat{\varepsilon}_i$.
2. Re-estimate equation (3) on the bootstrap sample, holding the regressors fixed at their original values, to obtain $\hat{\beta}_h^{*(b)}$.
3. Repeat steps 1–2 for $B = 999$ replications, and construct the $(1 - \alpha)$ confidence interval for β_h from the $\alpha/2$ and $1 - \alpha/2$ percentiles of the empirical distribution of $\{\hat{\beta}_h^{*(b)}\}_{b=1}^B$.

Multiplier distribution. The multiplier t is drawn from Mammen's (Mammen 1993) two-point distribution, which satisfies $E[t] = 0$, $\text{Var}(t) = 1$, and $E[t^3] = 1$. The third-moment condition is what distinguishes this choice from a simple Rademacher (± 1 with equal probability) sign flip, and ensures that the bootstrap approximates the sampling distribution of $\hat{\beta}_h$ to a higher order of accuracy when that distribution is itself skewed. Concretely, t takes the value $-\frac{\sqrt{5}-1}{2}$ with probability $\frac{\sqrt{5}+1}{2\sqrt{5}}$, and the value $\frac{\sqrt{5}+1}{2}$ with probability $\frac{\sqrt{5}-1}{2\sqrt{5}}$.

D Robustness Results for Emissions

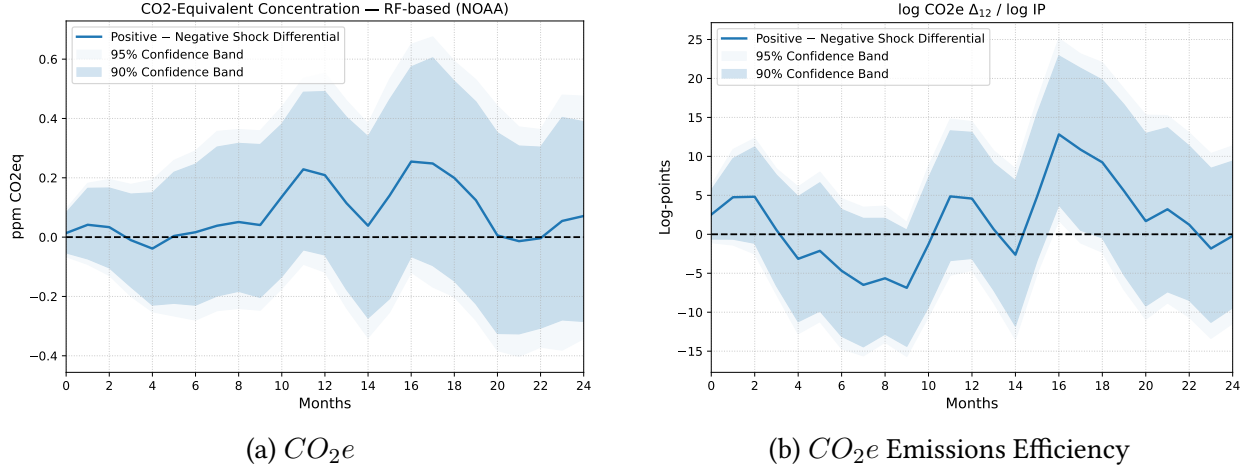
As described in subsection 2.4, I re-estimate equation (3) under four alternative specifications: no shock lags, 6 and 18 control and shock lags, and the shock trimmed to its 5th–95th percentiles. Figure D.1 plots the same robustness comparison for the remaining greenhouse gas series presented in figure 4: CO_2 , CH_4 , N_2O , and SF_6 , and confirms that the point estimates remain very similar.

Figure D.1: Robustness of Individual Greenhouse Gas Responses to Alternative Lag Specifications



Notes: Baseline impulse response (blue line, shaded 90%/95% confidence bands) to the oil supply news shock of Känzig (2021), overlaid with point estimates from four alternative specifications: no lags on the shock, 6 lags on both the target and the shock, 18 lags on both, and the shock trimmed to its 5th–95th percentiles. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Figure D.2: Differential Response to Positive versus Negative Oil Price Shocks



Notes: Point estimate and 90%/95% confidence bands for γ_h , estimated via equation (5). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024 in panel (a), and January 2003 – December 2024 in panel (b); outcome data extends to December 2025.

Asymmetric Response to Positive versus Negative Shocks To test for asymmetric dynamics, I augment equation (3) with an interaction between the shock and an indicator for a positive (price-increasing) realization:

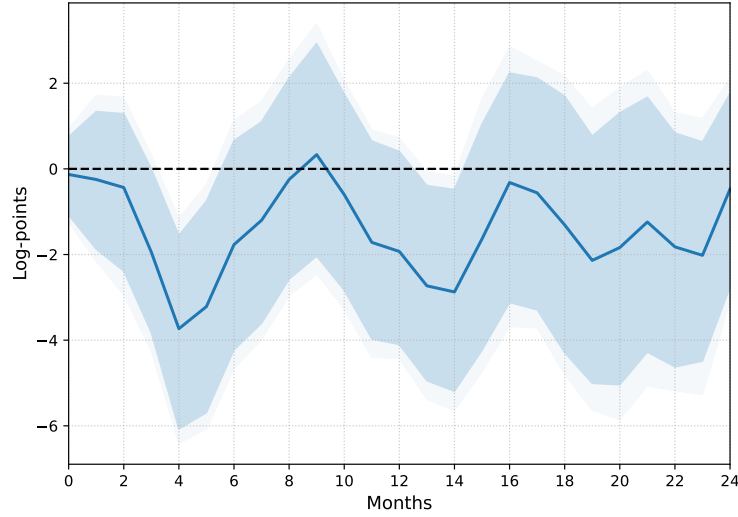
$$\Delta^{h+1}y_{t+h} = \alpha_h + \alpha_{m(t),h} + \beta_h \cdot s_t + \gamma_h \cdot s_t \times 1_{\{s_t > 0\}} + \sum_{k=1}^{12} (\rho_{h,k} y_{t-k} + \Gamma'_{h,k} X_{t-k} + \tau_{h,k} s_{t-k}) \quad (5)$$

where $1_{\{s_t > 0\}}$ equals 1 when the shock is positive (an oil price increase) and 0 otherwise. The coefficient γ_h is the differential slope for positive shocks relative to negative ones at horizon h : for a unit shock ($s_t = \pm 1$), the response to a positive shock is $\beta_h + \gamma_h$, while the response to a negative shock is $-\beta_h$.

Figure D.2 plots γ_h , the coefficient on the positive-shock interaction term in equation (5) and the differential slope between positive (price-increasing) and negative (price-decreasing) oil supply news shocks, for both CO_2e and emissions efficiency.

Emissions Efficiency Denominator Figure D.3 re-estimates the emissions-efficiency response of figure 3 using industrial production summed over the trailing 12 months, rather than its monthly level, as the denominator: $\log \left((\Delta^{12} CO_2e_t) / \sum_{j=0}^{11} IP_{t-j} \right)$. Both numerator and denominator are then annual flows covering the same 12 months, and the denominator is free of seasonality. The entire dynamics are very similar to those presented in panel b) of figure 3.

Figure D.3: Emissions Efficiency Response — 12-Month Industrial Production Denominator



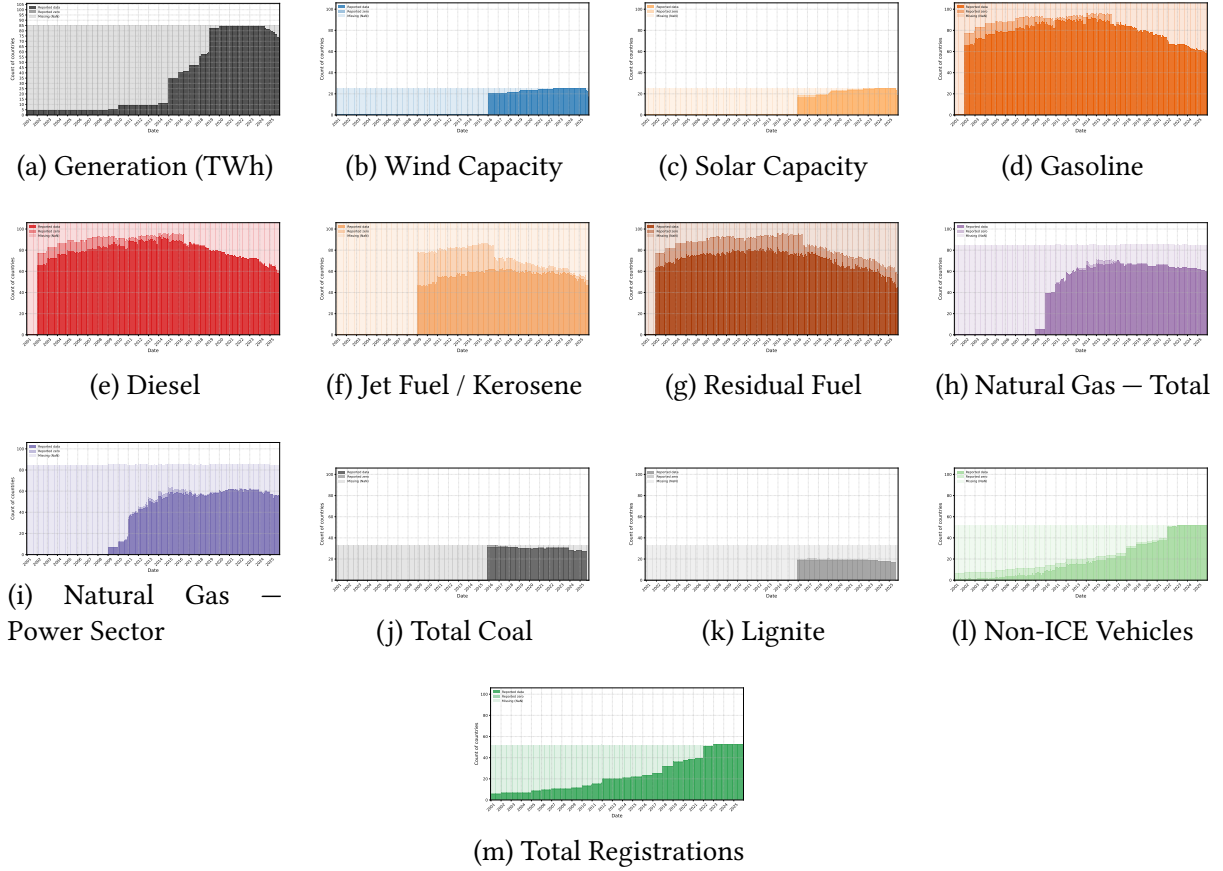
Notes: Impulse response to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in the log ratio of the 12-month difference in CO_2e concentration to the sum of the industrial production index over the same 12 months. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2003 – December 2024; outcome data extends to December 2025.

E Panel Data Description

Table E.1 lists every data source, with its frequency, sample span and number of non-missing observations, and, for panel datasets, the number of countries contributing at least 1 non-missing observation over the sample period. All samples are restricted to January 2001 onwards to coincide with the availability of the NOAA N_2O concentration series — the binding constraint on the emissions sample, since the shock and control series all predate 2001. Oil supply news shocks run through December 2024. Panel outcome data extends to December 2025; the additional year contributes only as forward outcomes y_{t+h} for shocks realised in 2024, leaving the shock identification sample unchanged. Table E.2 reports summary statistics for the time-series variables: the shocks, the global controls, and the outcomes.

Electricity generation data come from Ember, a monthly panel covering 85 countries from December 1998 onwards. Wind and solar installed capacity, also from Ember, is available for a much narrower sample of 25 countries starting in January 2016. Industrial production at the country level is sourced from the IMF, covering the OECD and selected emerging economies. Oil product consumption data (gasoline, diesel, jet fuel, LPG, and residual fuel) are from the Joint Organisations Data Initiative (JODI), as is natural gas consumption, both total and by the power sector. Coal production at the country level comes from the International Energy Agency

Figure E.1: Country Coverage Across Panel Data Sources



Notes: Each bar shows the number of countries with non-missing data in that month. Panel (a): electricity generation (Ember). Panels (b)–(c): wind and solar installed capacity (Ember). Panels (d)–(g): oil product consumption — gasoline, diesel, jet fuel/kerosene, and residual fuel (JODI). Panels (h)–(i): natural gas consumption, total and power sector (JODI). Panels (j)–(k): coal production, total and lignite (IEA). Panels (l)–(m): vehicle registrations, non-ICE and total. Each panel spans the full sample window, January 2001 – December 2025, regardless of each series’ own reporting start date.

(IEA), and new vehicle registrations from Andrew (2026). The nominal exchange rates used as a country-level control are from the IMF. Coverage across these datasets is unbalanced: the number of countries reporting data varies across variables and over time, as documented in figure E.1. Summary statistics for each variable used in the panel analysis, including the exchange rate, are reported in table E.3.

E.1 Ember–EIA Comparison

To validate the Ember dataset, I compare it against the annual international statistics published by the US Energy Information Administration (EIA). For each country and variable present in both datasets, I aggregate Ember’s monthly data to an annual total and compare it against the

Table E.1: Data Sources

Dataset	Source	Frequency	Countries	Start	End	Obs
GHG Concentrations (CO_2 , CH_4 , N_2O , SF_6)	NOAA	Monthly	—	2001	2025	300
Oil Supply News Shock	KZ	Monthly	—	2001	2024	288
Structural Oil Supply Shock	BH	Monthly	—	2001	2024	288
Industrial Production (OECD+6)	BH	Monthly	—	2001	2025	300
Oil and Gas Prices, CPI, Interest Rates	FRED	Monthly	—	2001	2025	299
Crude Oil Volatility Index (OVX)	FRED	Monthly	—	2007	2025	223
Global Financial Cycle	MR	Monthly	—	2001	2024	288
Financial and Macro Uncertainty	JLN	Monthly	—	2001	2025	300
Geopolitical Risk	CI	Monthly	—	2001	2025	300
Electricity Generation	Ember	Monthly	85	2001	2025	10093
Wind and Solar Capacity	Ember	Monthly	25	2016	2025	2754
Industrial Production	IMF	Monthly	55	2001	2025	15027
Vehicle Registrations	RA	Monthly	52	2001	2025	7110
Oil Products	JODI	Monthly	106	2002	2025	24085
Natural Gas	JODI	Monthly	80	2009	2025	12085
Coal Production	IEA	Quarterly	33	2016	2025	1188
Exchange Rates	IMF	Monthly	119	2001	2025	35542

Notes: Sample restricted to 2001–2025. COVID-19 (2020) is excluded from regressions but included in the counts above. Each row is described by one representative series: *Start*, *End* and *Obs*. count its non-missing observations, and for panel datasets *Countries* reports the number of countries with at least one non-missing observation. Summary statistics are reported in table E.2 (time series) and table E.3 (panels). KZ: Känzig (2021); BH: Baumeister and Hamilton (2019); MR: Miranda-Agrippino and Rey (2020); JLN: Jurado, Ludvigson, and Ng (2015); CI: Caldara and Iacoviello (2022); RA: Andrew (2026).

Table E.2: Time-Series Variables: Shocks, Controls and Outcomes

Variable	Source	Start	End	N	Mean	Std	Min	Median	Max
<i>Shocks</i>									
Oil Supply News Shock	KZ	2001	2024	288	0.01	0.64	-3.07	-0.01	1.93
Structural Oil Supply Shock	BH	2001	2024	288	0.07	0.63	-1.89	0.09	5.30
<i>Global Controls</i>									
Global Financial Cycle	MR	2001	2024	288	0.19	1.00	-2.92	0.07	2.59
Financial Uncertainty (3M-ahead)	JLN	2001	2025	300	0.96	0.15	0.72	0.96	1.43
<i>Outcomes</i>									
CO_2e Concentration (ppm)	NOAA	2001	2025	300	450.25	21.67	414.31	448.45	491.62
CO_2 Concentration (ppm)	NOAA	2001	2025	300	396.49	16.41	368.32	395.09	427.34
CH_4 Concentration (ppb)	NOAA	2001	2025	300	1831.17	53.95	1763.03	1814.40	1945.97
N_2O Concentration (ppb)	NOAA	2001	2025	300	326.54	6.88	316.11	325.93	339.65
SF_6 Concentration (ppt)	NOAA	2001	2025	300	8.13	2.25	4.69	7.91	12.49
Real WTI Oil Price (Jan 2010 = 1)	FRED	2001	2025	299	0.77	0.28	0.18	0.71	1.71
Real Henry Hub Gas Price (Jan 2010 = 1)	FRED	2001	2025	299	0.72	0.42	0.18	0.60	2.51
Industrial Production (OECD+6)	BH	2001	2025	300	85.01	14.16	60.03	85.13	110.67
10-Year Nominal Interest Rate (%)	FRED	2001	2025	300	3.21	1.19	0.62	3.36	5.39
10-Year – 1-Year Term Spread (pp)	FRED	2001	2025	300	1.28	1.19	-1.49	1.37	3.40
Macro Uncertainty (3M-ahead)	JLN	2001	2025	300	0.81	0.12	0.66	0.77	1.31
Geopolitical Risk	CI	2001	2025	300	110.09	51.27	46.90	96.09	512.53
Crude Oil Volatility Index (OVX)	FRED	2007	2025	223	38.36	15.94	16.68	35.47	163.65

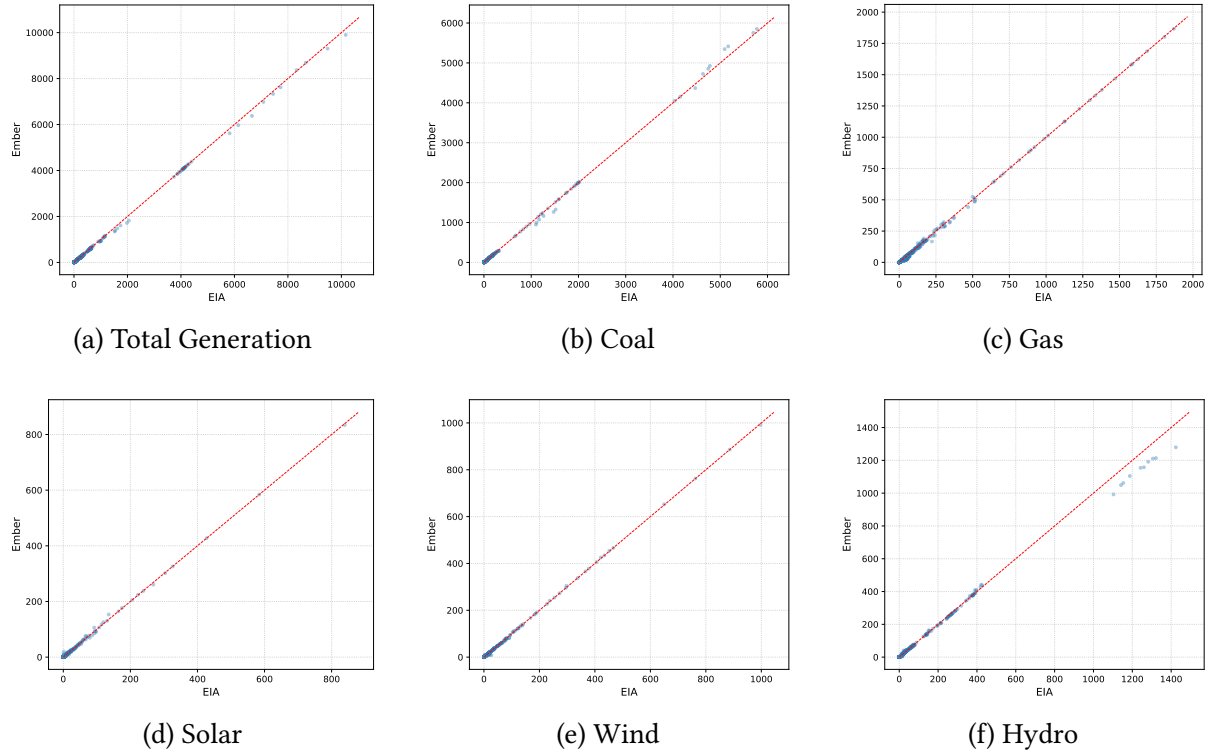
Notes: Sample restricted to years ≥ 2001 : *Start* is the later of that year and the series' first non-missing year, and *N*, Mean and all distributional statistics are computed over the same window. Oil supply shocks run through December 2024; outcome data extends to December 2025, with 2025 observations entering only as forward differences $y_{t+h} - y_{t-1}$ for shocks realised in 2024. Shocks and controls enter the right-hand side of every local projection; outcomes are the time-series dependent variables. KZ: Känzig (2021); BH: Baumeister and Hamilton (2019); MR: Miranda-Agrippino and Rey (2020); JLN: Jurado, Ludvigson, and Ng (2015); CI: Caldara and Iacoviello (2022); NOAA: Lan, Tans, and Thoning (2026) and Lan, Thoning, and Dlugokencky (2026).

Table E.3: Descriptive Statistics — Panel Variables

	Elec. Gen. (TWh)	Wind Cap. (GW)	Solar Cap. (GW)	IMF Ind. Prod.	ZEV Reg.	Alt+Hyb. Reg.	ICE Reg.	JODI Diesel	JODI Gasoline	JODI Jet Fuel	JODI LPG	JODI Res. Fuel	JODI Gas Power	JODI Gas Total	IEA Coal	IEA Steam & Cok. Coal	IEA Lignite	IMF Exch. Rate
Mean	30.47	28.92	41.84	106.41	7,084.42	10,632.07	116,201.74	8,067.21	7,659.78	1,691.23	2,495.50	2,067.93	1,090.12	2,922.50	65,121.02	75,030.92	7,953.47	1,442.96
Std. Dev.	94.70	69.24	120.32	24.81	44,728.54	33,234.28	291,979.26	17,668.08	31,014.59	5,670.42	6,992.32	4,103.06	3,469.28	8,879.28	183,686.96	202,968.77	8,557.78	47,727.91
Min	0.02	0.19	0.00	32.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.12
p25	1.53	3.34	1.77	93.51	16.00	96.25	6,992.25	599.20	321.10	15.68	81.20	51.60	52.00	246.00	620.00	240.50	1,016.00	2.00
Median	4.91	6.29	11.22	103.71	227.00	1,232.00	21,474.00	2,461.80	1,157.65	310.00	371.20	387.00	272.00	594.00	7,768.00	2,257.50	6,348.00	8.81
p75	17.53	22.98	35.00	115.00	2,064.75	6,309.50	91,977.75	7,195.69	4,394.00	1,150.48	1,960.56	2,031.75	763.00	2,856.00	31,914.50	66,272.00	11,520.00	104.72
Max	1,015.19	640.01	1,442.08	450.70	829,100.00	555,317.00	2,665,499.00	140,150.00	306,022.00	58,116.00	89,989.88	43,249.69	43,258.00	111,046.00	1,283,191.00	1,283,191.00	45,747.00	4,087,157.94
N	10,093	2,751	2,674	15,027	7,110	7,110	7,110	24,291	24,085	14,763	24,076	24,137	10,497	12,085	1,188	950	765	35,542
N Countries	85	25	25	55	52	52	52	106	106	94	106	106	81	80	33	30	22	119
N Periods	300	120	120	300	300	300	300	288	288	204	288	288	204	204	39	39	39	300
Max ctry./period	84	25	25	55	52	52	52	96	96	87	96	96	64	71	33	29	21	119
Min ctry./period	4	17	18	35	6	6	6	60	60	51	59	57	7	5	27	20	17	109
Start date	2001-01	2016-01	2016-01	2001-01	2001-01	2001-01	2001-01	2002-01	2002-01	2009-01	2002-01	2002-01	2009-01	2009-01	2016-01	2016-01	2016-01	2001-01
End date	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-07	2025-07	2025-07	2025-12

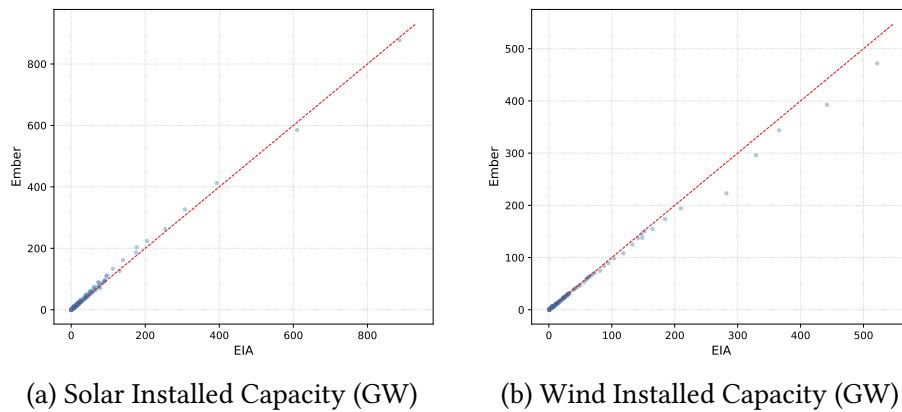
Notes: Elec. = Electricity; Cap. = Capacity (solar in GW AC or GW DC, as reported by each country); IMF Ind. Prod. = IMF Industrial Production Index; ZEV = Zero-Emission Vehicle; ICE = Internal Combustion Engine; Alt+Hyb. = Alternative Fuel + Hybrid; Cok. = Coking; Res. = Residual; JODI = Joint Organisations Data Initiative (quantities in KBBL = thousands of barrels, except gas power and total in M3 - cubic meters); IEA = International Energy Agency (quantities in KT = kilotons); Ctry. = Countries; IMF Exch. Rate = IMF nominal exchange rate, in domestic currency per US dollar. All the data are reported at a monthly frequency except for the IEA coal data which has a quarterly frequency.

Figure E.2: Ember vs. EIA Annual Electricity Generation



Notes: Each point represents 1 country-year observation. Ember monthly generation is aggregated to annual totals and compared against EIA annual reporting. Both axes in TWh.

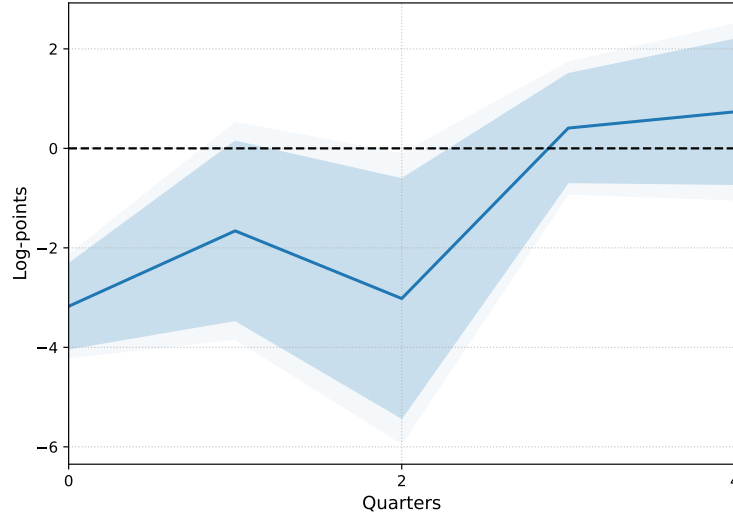
Figure E.3: Ember vs. EIA Annual Installed Capacity



Notes: Each point represents 1 country-year observation. Ember annual capacity is compared against EIA annual reporting. Both axes in GW.

corresponding EIA annual figure. Figure E.2 plots this comparison for electricity generation by fuel type, and figure E.3 extends the comparison to installed wind and solar capacity.

Figure F.1: Response of Total Coal Production to an Oil Supply News Shock



Notes: Impulse response to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log quarterly coal production; the horizon is measured in quarters. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-quarter. Regressions are weighted by the lag of the raw production level. COVID-19 (2020) is excluded from the sample. Shocks: January 2017 – December 2024; outcome data extends to July 2025.

F Panel Analysis Further Results

F.1 Total Coal Production

Figure 8 in the main text separates coal production by type. Figure F.1 reports the same specification for total coal production (steam, coking, and lignite combined), which decreases by 3.18 percentage points on impact and remains depressed for roughly half a year before recovering, following the dynamics of steam and coking coal.

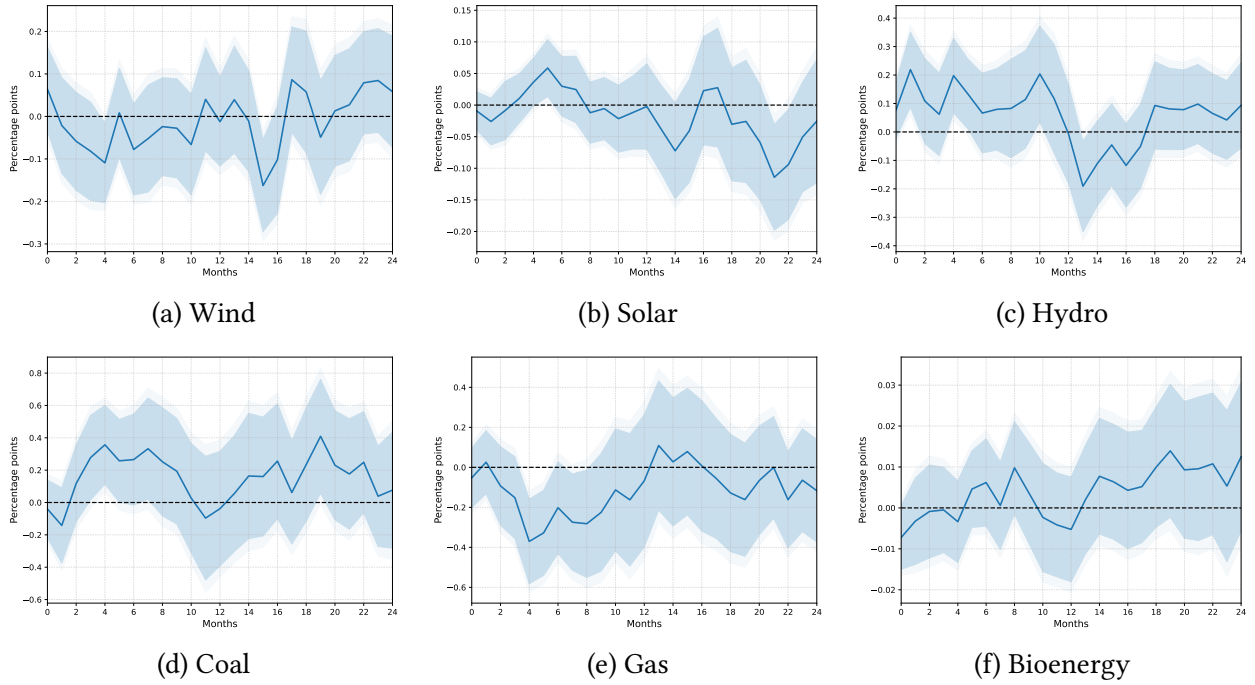
F.2 Electricity Generation as a Share of Total

Figure F.2 reports wind, solar, hydro, coal, gas, and bioenergy generation as a percentage of total electricity generation, complementing the TWh-based responses reported in figures 8, 9 and 12.

F.3 Ember Capacity Data

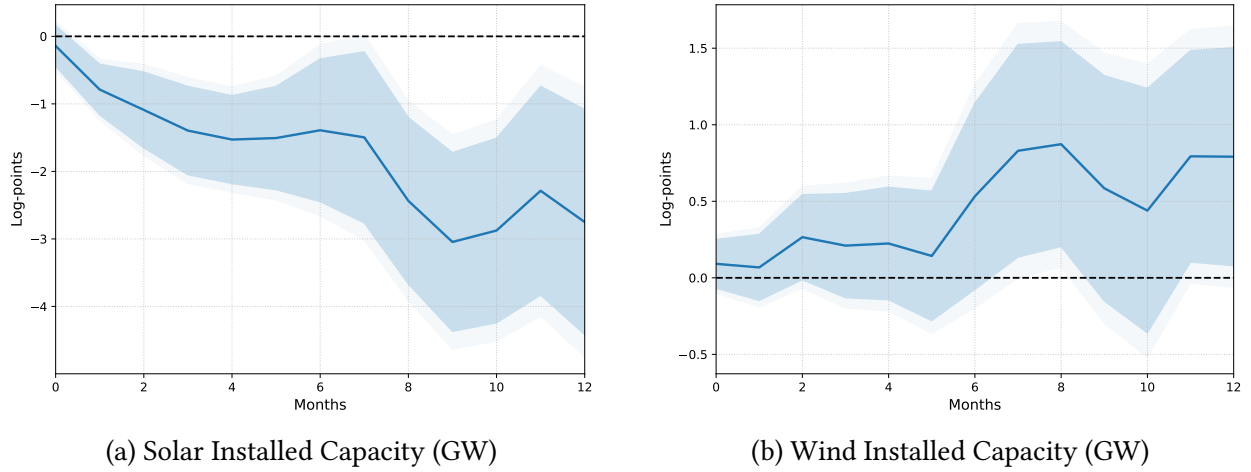
The capacity panel covers 25 countries starting in January 2016 — a sample that is both geographically narrower and temporally shorter than the generation panel, which draws on 85 countries from January 2001. I present the impulse response functions for solar and wind nameplate capacity in figure F.3. As with coal, I consider only a 12-month horizon since the sample

Figure F.2: Electricity Generation by Source as a Percentage of Total Generation



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in the respective source's percentage share of total monthly electricity generation. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of total electricity generation (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Figure F.3: Response of Renewable Installed Capacity to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly installed renewable capacity (GW). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective installed capacity (GW). The horizon is capped at 12 months. Capacity data cover 25 countries. COVID-19 (2020) is excluded from the sample. Shocks: January 2017 – December 2024; outcome data extends to December 2025.

is much shorter, for most countries starting only in 2016. On top of being shorter, the capacity panel is also narrower: its fewer countries account for a smaller share of total wind and solar generation than the generation panel does.

Any direct comparison between generation and capacity impulse responses risks conflating genuine causal effects with artefacts of sample composition: the 25 countries with capacity data may have systematically different generation dynamics, independent of any link between capacity and generation.

To address this concern, I re-estimate the generation local projections restricting the sample to those country-month observations for which the corresponding installed capacity variable is non-missing. Specifically, solar generation is estimated on the subsample defined by non-missing solar capacity, and wind generation on the subsample defined by non-missing wind capacity. This yields impulse responses on an estimation sample identical to those used for the capacity results in figure F.3, permitting a like-for-like comparison. The results are presented in figure F.4.

For solar, the dynamics closely replicate those from the full sample: the response follows the same trajectory stabilizing at -5.62 log-points at the 2-year horizon, compared to -5.05 log-points in the full sample. The solar generation decline is therefore not driven by differences in country coverage between the generation and capacity panels. For wind, the restricted sample produces a volatile response: generation initially declines but subsequently rises, with wide confidence bands throughout. This noisy pattern probably reflects the substantially smaller sample and the

inability to capture the underlying signal. It also plausibly explains why wind capacity itself increases following the shock in figure F.3: on this same narrow set of countries, generation — not just capacity — also rises, suggesting the pattern reflects the composition of the capacity sample rather than a genuine investment response. The capacity data also have two further limitations worth noting: they do not account for depreciation, and they record only nameplate capacity, which may not reflect the actual potential power output of the installed capital.

F.4 Vehicle Registrations

Total Registrations Figure 10 in the main text reports ICE and non-ICE vehicle registrations separately. Figure F.5 reports the same specification for total registrations (ICE and non-ICE combined). Total registrations fall by 1.99 percentage points at their trough, 8 months after the shock, and remain depressed after 2 years, 1.63 percentage points below trend.

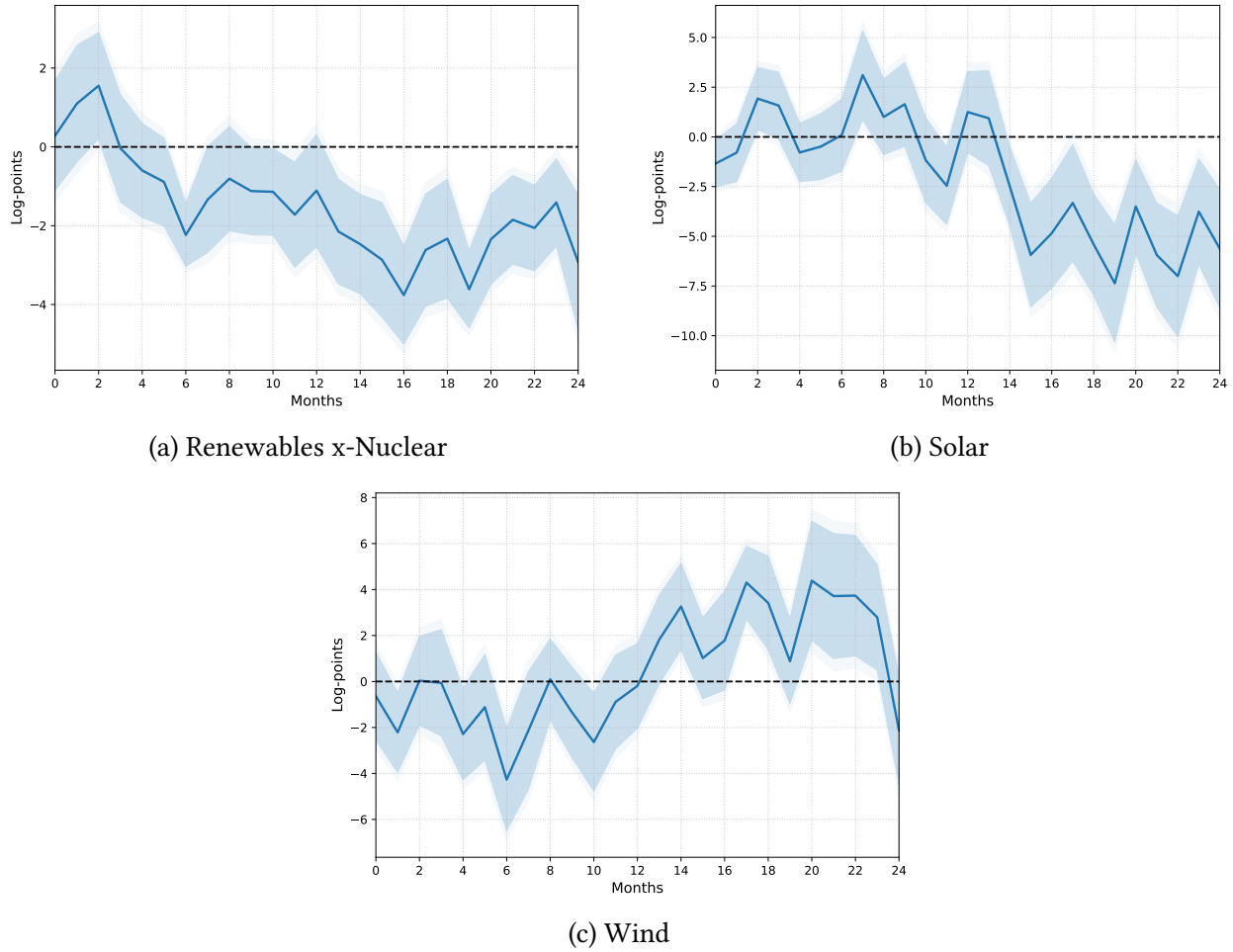
Non-ICE Registrations Figure F.6 decomposes non-ICE registrations into zero-emission vehicles (ZEV), hybrids, and alternative-fuel ICE vehicles (ICE-Alt), which run on LPG or ethanol only. Flex-fuel cars can run entirely on petrol and are therefore counted as ICE. ZEV and hybrid registrations drive the aggregate pattern in figure 10: ZEV registrations fall 7.40 percentage points after 2 months and overshoot to 8.12 percentage points at month 15, while hybrids fall 4.54 percentage points before overshooting to 2.79 percentage points. ICE-Alt registrations barely fall, by only 0.96 percentage points, but overshoot earlier, reaching 6.44 percentage points after 10 months.

F.5 Panel Robustness Checks

Every panel local projection reported in the main text (sections 3 and 4) is re-estimated under four alternative specifications, each targeting a distinct potential concern and each overlaid as a point estimate on the corresponding baseline figure:

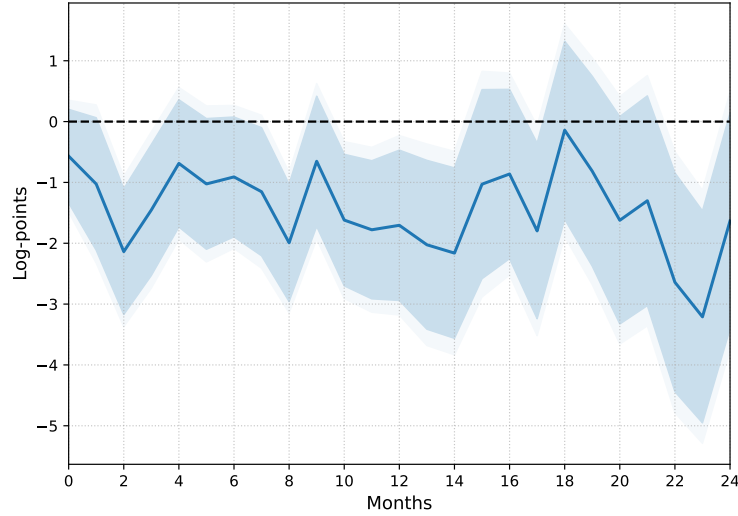
- *No lags for s_t* : drops the shock’s own lags from the right-hand side of equation (4), to show that the results are not purely driven by noise.
- *Country + Month FE*: replaces the baseline’s country-month fixed effects, which allow each country its own seasonal pattern, with separate, additive country and calendar-month fixed effects, imposing a common seasonal pattern across countries.
- *Alternative Weight (12-m Rol. Avg)*: replaces the baseline’s single-period lagged weight with a 12-month rolling average of the same weight variable, to decrease the effect of seasonality in the weighting scheme itself, at the expense of potentially blurring genuine within-year changes in the true cross-country distribution of energy activity.

Figure F.4: Response of Renewable Electricity Generation to an Oil Supply News Shock — Capacity Sample



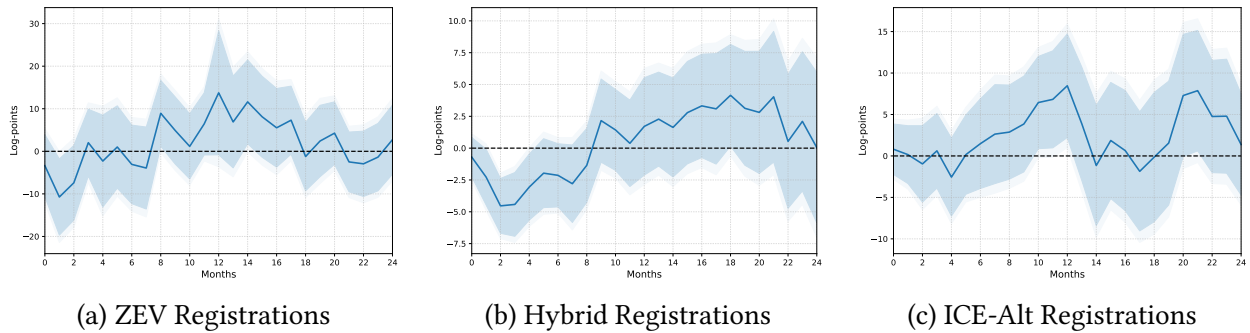
Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly electricity generation (TWh) from the respective source. Each regression is restricted to country-month observations for which the corresponding installed capacity variable is non-missing: solar generation and renewables-excluding-nuclear on the solar capacity sample, wind generation on the wind capacity sample. This matches the estimation sample to the capacity regressions reported in figure F.3. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective generation series (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2017 – December 2024; outcome data extends to December 2025.

Figure F.5: Response of Total Vehicle Registrations to an Oil Supply News Shock



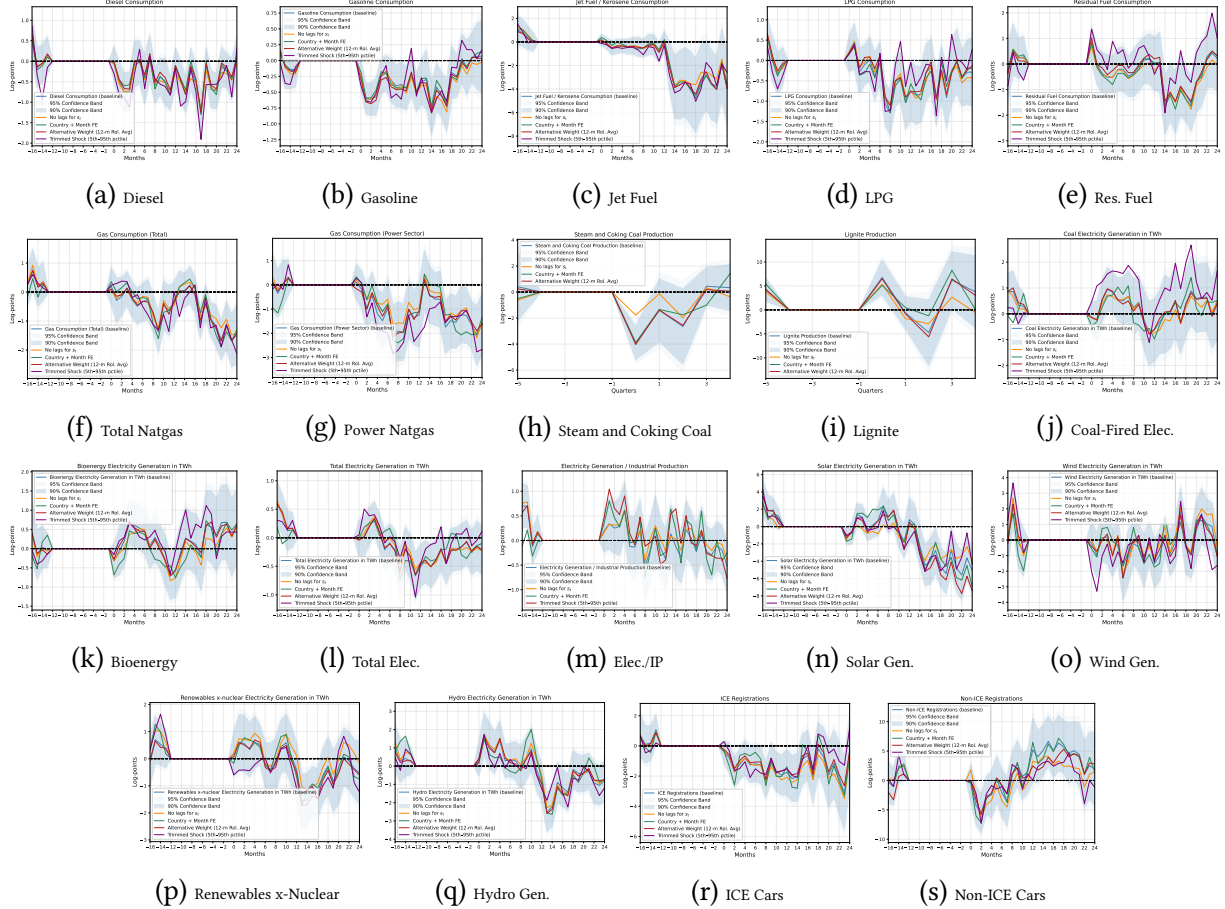
Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly vehicle registrations. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective vehicle registration count. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Figure F.6: Response of Non-ICE Vehicle Registrations by Type to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly vehicle registrations. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective registration count. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Figure F.7: Robustness of All Panel Local Projections to Alternative Specifications



Notes: Baseline impulse response (blue line, shaded 90%/95% confidence bands) to the oil supply news shock of Känzig (2021), overlaid with point estimates from the four alternative specifications described above, including the 16 months (5 quarters for coal) preceding the shock, for every panel regression reported in the main text: refined oil consumption (figure 6), natural gas consumption (figure 7), coal production and coal-fired electricity generation (figure 8), total electricity generation and electricity-to-industrial-production (figure 11), bioenergy generation (figure 9), renewable generation by source (figure 12), and vehicle registrations (figure 10). COVID-19 (2020) is excluded from the sample.

- *Trimmed Shock (5th–95th pctile):* re-estimates using the oil news shock with values outside its own 5th–95th percentile band (computed over 2001–2025) set to missing, to ensure that no outlier shock realisations are driving the estimated dynamics.

Figure F.7 plots all four overlays for every panel series reported in the main text, together with the pre-trends — the responses in the 16 months (5 quarters for coal) preceding the shock — with one exception: the *trimmed-shock* check is omitted for coal (steam/coking coal and lignite), whose sample is far smaller than the other panels (33 countries, mostly entering between 2016 and 2016), and would imply even fewer observations.

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